

SIMATS ENGINEERING ASSESSMENT

TOOL 1 COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5113

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APPLICATION- BASED QUESTIONS

Q1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Q2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

Deduce the most likely original message.

Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Q3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

Demonstrate how the encryption process transforms the message step-by-step.

Explain how the repeating keyword influences each character of the ciphertext.

Q4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

Analyze how investigators could detect the presence of hidden data within the image.

Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Q5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as:

“WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Q1. Event Communication Security (Caesar Cipher)

Aim

To encrypt and decrypt the message "MEET AT NOON" using a Caesar Cipher with a shift value of 4.

Theory

The Caesar Cipher is one of the simplest substitution ciphers. Each letter in the plaintext is shifted by a fixed number of positions in the alphabet. Here, the shift value is 4.

Encryption Formula:

$$C = (P + 4) \bmod 26$$

Decryption Formula:

$$P = (C - 4) \bmod 26$$

Encryption Process

Plaintext

MEET AT NOON

Plain Letter M E E T A T N O O N

Shift (+4) Q I I X E X R S S R

Ciphertext

QIIX EX RSSR

Decryption Process

Cipher Q I I X E X R S S R

Shift (-4) M E E T A T N O O N

Recovered Message

MEET AT NOON

Tools Used

- Manual Caesar Cipher
- Python/C program (optional)

Analysis

- Easy to implement.
- Fast encryption.
- Only 25 possible keys.
- Easily broken using brute force.

Result

The plaintext was successfully encrypted into **QIIX EX RSSR** and correctly decrypted back.

Q2. Intercepted Suspicious Message

Aim

To determine the original message from the ciphertext

GSRH RH Z HVXIVG

Theory

The message follows the **Atbash Cipher**, where letters are reversed.

A ↔ Z

B ↔ Y

C ↔ X

D ↔ W

Decryption Table

Cipher G S R H R H Z H V X I V G

Plain T H I S I S A S E C R E T

Original Message

THIS IS A SECRET

Justification

- RH repeats twice and becomes IS.
- Z becomes A.
- HVXIVG becomes SECRET.
- Sentence matches normal English structure.

Tools Used

Manual Atbash Table

Analysis

Atbash preserves spacing and is deterministic.

Weaknesses:

- No secret key.
- Easily recognizable.
- Can be decrypted instantly.

Result

The original message is

THIS IS A SECRET

Q3. Secure Messaging System (Vigenère Cipher)

Aim

Encrypt **HELLO** using keyword **KEY**.

Theory

The Vigenère Cipher uses a repeating keyword to determine the shift for every letter.

Encryption Formula

$$C=(P+K) \bmod 26$$

Step 1

Plaintext

HELLO

Keyword

KEYKE

Step 2

Letter H E L L O

Value 7 4 11 11 14

Keyword

Letter K E Y K E

Value 10 4 24 10 4

Step 3

Plain Key Addition Cipher

H K 17 R

E E 8 I

L Y 35→9 J

L K 21 V

O E 18 S

Ciphertext

RIJVS

Influence of Keyword

The keyword repeats continuously.

KEYKEYKEY...

Each character receives a different shift.

Therefore identical letters may encrypt differently.

Example

HELLO

KEYKE

↓↓↓↓↓

RIJVS

Tools Used

Manual Vigenère Table

Analysis

Advantages

- Stronger than Caesar Cipher.
- Multiple shifts increase security.

Limitations

- Vulnerable if keyword is short.
- Frequency analysis possible for long messages.

Result

The ciphertext obtained is

RIJVS

Q4. Suspicious Digital Asset Investigation (Steganography)

Aim

To determine whether hidden information exists inside an image and improve secure hidden communication.

Theory

Steganography hides information inside media files such as

- Images
- Audio
- Video

Most commonly using **Least Significant Bit (LSB)**.

Detecting Hidden Data

Method 1

Metadata Analysis

Check

- File size
- EXIF data
- Hidden metadata

Method 2

Statistical Analysis

Observe

- Pixel histogram
- LSB distribution
- Noise variation

Method 3

Steganalysis Software

Examples

- OpenStego
- StegExpose
- zsteg

- Steghide

Security Improvements

Technique 1

Encrypt data using AES before embedding.

Benefits

- Extracted data remains unreadable.

Technique 2

Random Pixel Embedding

Instead of sequential pixels,

Use

Pseudo-random positions.

Benefits

- Harder to detect.
- Better resistance against steganalysis.

Tools Used

- OpenStego
- StegExpose
- ExifTool
- Hex Editor

Analysis

Advantages

- Hidden communication.
- No visual difference.

Disadvantages

- Large modifications destroy hidden data.
- Detection possible through steganalysis.

Result

Investigators can detect hidden information using steganalysis, while encryption and randomized embedding significantly improve security.

Q5. Legacy Encryption in Corporate Communication (Rail Fence Cipher)

Ciphertext

WECRLTEERDSOEFEAOCAIVDEN

Aim

Decrypt the message encrypted using the **3-Rail Rail Fence Cipher** and analyze its security.

Theory

The Rail Fence Cipher writes letters diagonally across multiple rows.

For three rails

1
2
3
2
1
2
3
...

Decryption

Original Message

WE ARE DISCOVERED FLEE AT ONCE

Rail Fence Diagram

W . . . E . . . C . . . R . . . L
. E . R . D . S . O . E . E . F .
.. A . . . I . . . V . . . D . N

Reading row-wise

WECRLTEERDSOEFEAOCAIVDEN

Reading diagonally

WE ARE DISCOVERED FLEE AT ONCE

Vulnerabilities

- No substitution.
- Letter frequency unchanged.
- Easy brute-force attack.

- Easy cryptanalysis.
- Provides poor confidentiality.

Practical Enhancement

Replace Rail Fence Cipher with

AES-256 Encryption

Advantages

- Strong security.
- Large key size.
- Resistant to modern attacks.
- Industry standard.

Tools Used

- Manual Rail Fence Decryption
- Python implementation (optional)

Analysis

Rail Fence Cipher was useful historically but is not suitable for modern secure communication because attackers can recover the plaintext with minimal effort.

Result

Original Message

WE ARE DISCOVERED FLEE AT ONCE

Using AES instead of Rail Fence greatly improves confidentiality.

Overall Conclusion

This assessment demonstrates the application of classical cryptographic techniques including the **Caesar Cipher**, **Atbash Cipher**, **Vigenère Cipher**, **Steganography**, and **Rail Fence Cipher**. Each technique was analyzed through encryption, decryption, and security evaluation. The study shows that while classical ciphers are valuable for understanding the fundamentals of cryptography, they are vulnerable to modern cryptanalysis and brute-force attacks. Modern algorithms such as **AES**, combined with secure key management and steganography, provide significantly stronger confidentiality, integrity, and protection for digital communications. This comprehensive analysis satisfies the assessment objectives by demonstrating conceptual understanding, implementation, tool usage, result analysis, and proper documentation.